FOREST NETRA

Adafruit IO Compact Data Format & Dashboard Reference

1. System Rule

IMPORTANT: The existing Environmental Node program and Serial Monitor output must NOT be changed.

The current system already successfully:

- Reads Environmental-node sensors.
- Receives Tree-node data.
- Receives Soil-node data.
- Displays the complete Forest Netra dataset on the Serial Monitor.

Adafruit IO is an **additional cloud layer**.

The existing data flow remains:

Environmental sensors + Tree data + Soil data
↓
Environmental ESP8266
↓
Existing Serial Monitor output — unchanged
↓
Additional Adafruit IO publishing
↓
One compact message
↓
Adafruit IO dashboard

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2. Adafruit IO Architecture

Only **one Adafruit IO feed** will be used for the Forest Netra sensor dataset.

Feed name:

forest-netra-data

The Environmental ESP8266 will periodically publish **one compact message** containing the latest complete numerical reading.

Example:

T=27.54,P=1003.63,A=80.41,H=77.00,L=305,C=655,G=0.1084,M=0.00,ST=25.90,E=0,PH=7.00,N=0,PP=0,K=0

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3. Compact Message Format

Format:

KEY=VALUE,KEY=VALUE,KEY=VALUE,...

Separators:

, → separates parameters

= → separates parameter name from its value

Example:

T=27.54,P=1003.63,A=80.41,H=77.00,L=305,C=655,G=0.1084,M=0.00,ST=25.90,E=0,PH=7.00,N=0,PP=0,K=0

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4. Environmental Parameters

Code	Parameter	Source	Unit
T	BMP390 Temperature	BMP390	°C
P	Atmospheric Pressure	BMP390	hPa
A	Altitude	BMP390	m
H	Relative Humidity	DHT11	%RH
L	Light Intensity	BH1750	lux

Example:

T=27.54

P=1003.63

A=80.41

H=77.00

L=305

These values are sensor measurements and do not currently have Forest Netra classification thresholds in the Tree-node code.

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5. Tree Parameters

Code	Parameter	Unit
C	Clear Intensity	raw sensor value
G	GLI	dimensionless

Example:

C=655

G=0.1084

5.1 GLI Calculation

The Tree node calculates:

r = Raw Red / Raw Clear

g = Raw Green / Raw Clear

b = Raw Blue / Raw Clear

GLI = ((2 × g) - r - b) / ((2 × g) + r + b)

The resulting value is represented by:

G

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6. Tree Health Classification

The current Tree-node program uses the following exact thresholds.

Plant Status

GLI >= 0.15

→ Healthy

0.08 <= GLI < 0.15

→ Moderate (Stressed/Deficient)

GLI < 0.08

→ Diseased/Chlorotic

Therefore:

GLI (G)	Plant Status
>= 0.15	Healthy
0.08 to < 0.15	Moderate (Stressed/Deficient)
< 0.08	Diseased/Chlorotic

Example:

G=0.1084

corresponds to:

Plant Status: Moderate (Stressed/Deficient)

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7. Tree Environment Classification

The current Tree-node program uses **Clear Intensity (C)**.

C < 50

→ Unstable Light

C > 10000

→ Unstable Light

50 <= C <= 10000

→ Stable Light

Therefore:

Clear Intensity (C)	**Environment**
< 50	Unstable Light
50–10000	Stable Light
> 10000	Unstable Light

Example:

C=655

corresponds to:

Environment: Stable Light

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8. Tree Reflectance Classification

The current program uses:

C > 300

→ High

C <= 300

→ Low

Therefore:

Clear Intensity (C)	**Reflectance**
> 300	High
<= 300	Low

Example:

C=655

corresponds to:

Reflectance: High

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9. Tree Nutrient Classification

The current program uses both GLI and the relative RGB values.

High

GLI >= 0.15

AND

Green > Red

AND

Green > Blue

→ High (Nitrogen Rich)

Medium

GLI >= 0.00

→ Medium

This condition is reached when the High condition is not satisfied but GLI is still non-negative.

Low

GLI < 0.00

→ Low (Deficient)

Therefore:

Condition	Nutrient Status
G >= 0.15 AND Green > Red AND Green > Blue	High (Nitrogen Rich)
0.00 <= G < 0.15	Medium
G < 0.00	Low (Deficient)

Important: These are the thresholds/logic currently implemented in the prototype. They should not be presented as scientifically validated nutrient concentration thresholds.

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10. Soil Parameters

Code	Parameter	Unit
M	Soil Moisture	%
ST	Soil Temperature	°C
E	Electrical Conductivity	µS/cm
PH	Soil pH	pH
N	Nitrogen	mg/kg
PP	Phosphorus	mg/kg
K	Potassium	mg/kg

Example:

M=0.00

ST=25.90

E=0

PH=7.00

N=0

PP=0

K=0

The current lab reading of zero for some soil parameters is not being treated as a fault; the sensor is currently being tested outside the intended soil measurement condition.

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11. Complete Example

Compact message sent to Adafruit IO

T=27.54,P=1003.63,A=80.41,H=77.00,L=305,C=655,G=0.1084,M=0.00,ST=25.90,E=0,PH=7.00,N=0,PP=0,K=0

Dashboard interpretation

T = 27.54 °C

P = 1003.63 hPa

A = 80.41 m

H = 77.00 %RH

L = 305 lux

C = 655

G = 0.1084

M = 0.00 %

ST = 25.90 °C

E = 0 µS/cm

PH = 7.00

N = 0 mg/kg

PP = 0 mg/kg

K = 0 mg/kg

Using the current Tree-node thresholds:

G = 0.1084

→ Plant Status = Moderate (Stressed/Deficient)

C = 655

→ Environment = Stable Light

C = 655

→ Reflectance = High

G = 0.1084

→ Nutrient Status = Medium

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12. Dashboard Display Requirements

The future dashboard should be able to display:

Environmental

Temperature 27.54 °C

Pressure 1003.63 hPa

Altitude 80.41 m

Humidity 77.00 %RH

Light 305 lux

Tree

Clear Intensity 655

GLI 0.1084

Plant Status Moderate (Stressed/Deficient)

Reflectance High

Environment Stable Light

Nutrient Status Medium

Soil

Moisture 0.00 %

Temperature 25.90 °C

EC 0 µS/cm

pH 7.00

Nitrogen 0 mg/kg

Phosphorus 0 mg/kg

Potassium 0 mg/kg

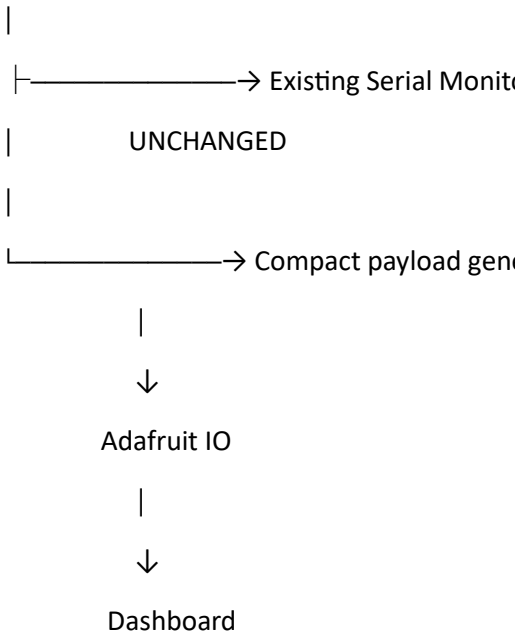
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13. Critical Implementation Rule

The existing Serial Monitor output is **not to be redesigned or replaced**.

The Adafruit IO implementation must work like this:

Existing sensor/data-processing code



The cloud code should use the **same variables already being used by the working program**.

It should NOT create a second independent sensor-reading system.

This prevents the Serial Monitor and cloud dashboard from accidentally displaying different readings.

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14. Future Cloud Message

The intended cloud message remains:

T=27.54,P=1003.63,A=80.41,H=77.00,L=305,C=655,G=0.1084,M=0.00,ST=25.90,E=0,PH=7.00,N=0,PP=0,K=0

One message.

One Adafruit IO feed.

All current numerical data.

The dashboard will later parse this message and display the individual values and calculated Tree status labels.